

Capstone Slide Deck

Slide Deck — KijaniKiosk Capstone (Track A)

Speaker notes in *italics*; export this file to PDF for submission (`./scripts/md-to-pdf.sh docs/slide-deck.md "Capstone Slide Deck"`, or a proper slide tool if you want real slide breaks for the live presentation).

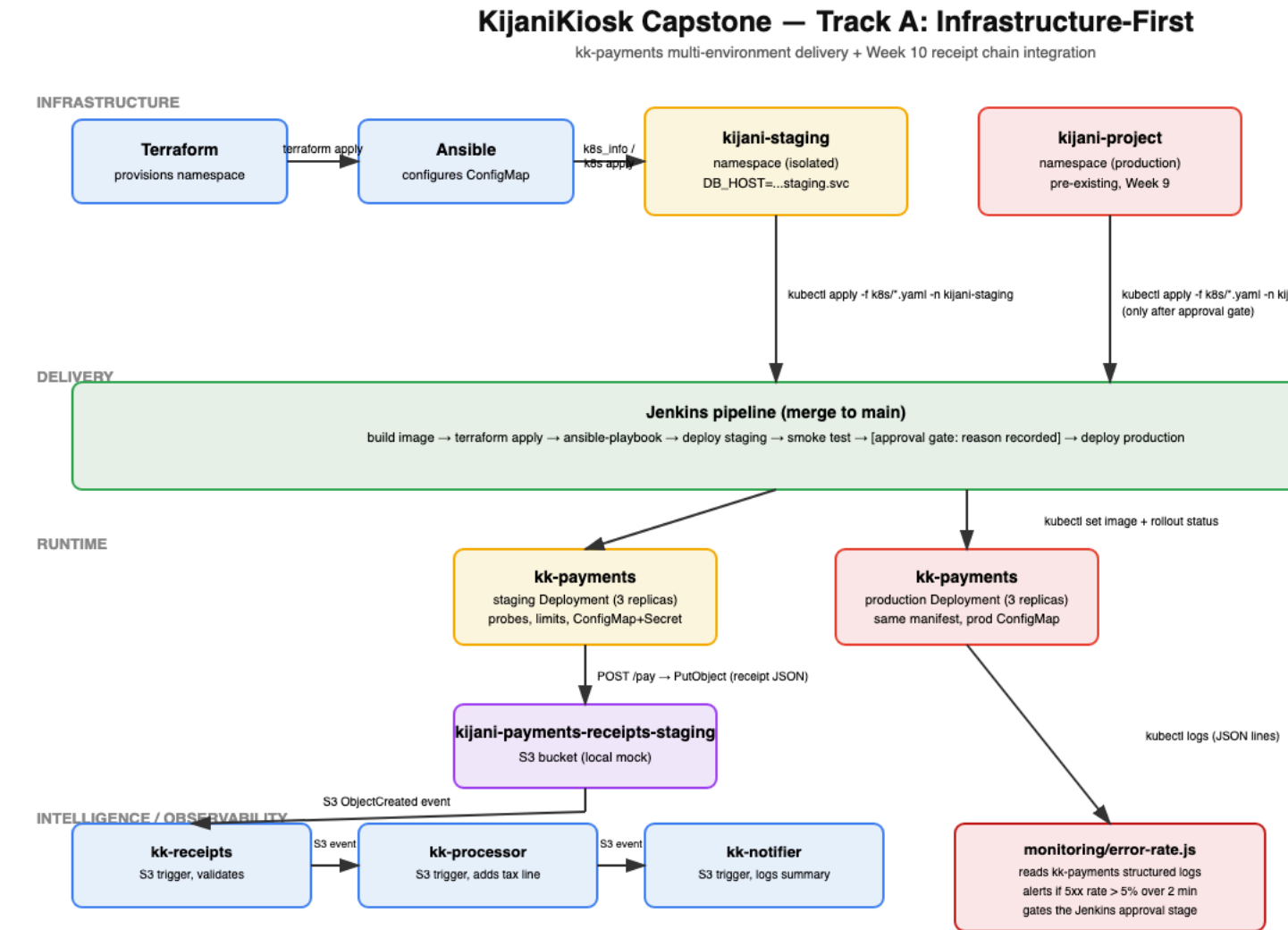
1. Title

KijaniKiosk Capstone — Track A: Infrastructure-First [Your name] · [Date] · github.com/mishobo/kijanikiosk-devops-foundation

2. Problem

kk-payments deploys to a single environment with no isolation between testing and production, and the Week 10 serverless receipt chain has never actually been fired by a real payment. *One paragraph, matches docs/scope.md problem statement verbatim.*

3. Architecture



4. Key Decision

I chose a shared local S3 mock directory (`/tmp/kijani-s3-local`) over a real AWS bucket for the receipt chain integration, because it requires no cloud account for local development and keeps the demo reproducible on a laptop with no network dependency. The trade-off: on minikube's docker driver, the Pod's filesystem and the laptop's are not the same `/tmp`, which is a real seam failure documented in `docs/runbook.md` and fixed with `minikube mount`. In production this would be a real S3 bucket with IAM roles, and the seam would move from "shared `/tmp` directory" to "correctly scoped IAM policy" — a different, more standard problem.

5. AI Tooling

Built with Claude (Claude Code CLI) as the primary author of first-draft code across Terraform, Ansible, the Jenkinsfile, the serverless handlers, and this documentation. - **Right**: correctly identified the minikube-docker-daemon build target convention and the ConfigMap/bucket-name DRY pattern project.md calls out. - **Wrong**: shipped a broken conditional in the Jenkinsfile's build stage and a `hostPath` assumption that silently breaks on minikube's docker driver — both caught and fixed/documented in `docs/ai-governance-log.md`. - **Changed**: rewrote the Jenkinsfile build stage as a single shell block; added the runbook's "integration seam that breaks first" section instead of silently shipping the `hostPath` assumption.

6. Production Gaps

1. **No real S3/IAM** — `serverless.yml`'s `resources` block declares real AWS S3 buckets but nothing has run `serverless deploy` against a real account; remediation: provision via CI with a scoped IAM role restricted to the three named buckets (already least-privilege in the `iam.role.statements` block).
 2. **No staging database** — `DB_HOST` in the staging ConfigMap has no backing Postgres instance; remediation: add a `kk-postgres StatefulSet` or managed RDS instance per environment before `kk-payments` makes real DB calls.
 3. **Jenkinsfile not exercised against a live Jenkins controller in this repo's own CI** — remediation: stand up a Jenkins instance (Docker Compose is sufficient for a demo) and point a Pipeline job at this repo before the next iteration.
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7. Governance Checklist (optional depth slide)

`docs/governance-checklist.md` — six controls (secrets, least privilege, idempotency, blast radius, drift risk, human review evidence) applied to every AI-generated file in this repo; see `docs/ai-governance-log.md` for which controls caught which issues.